ERRORQUAKE: Heavy-Tailed Error Severity Distributions in Open-Weight Large Language Models

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Abstract

At matched accuracy, open-weight LLMs differ substantially in the shape of their error severity distribution — a difference invisible to the scalar error rate. Hallucination benchmarks report a single error count and treat all errors as equivalent, yet a wrong date and a fabricated court ruling differ by orders of magnitude. We introduce ERRORQUAKE-10K, a 10,000-query benchmark scoring each response on a continuous 0–4 severity scale across 8 domains and 5 difficulty tiers, and we fit per-model severity distributions for 21 open-weight models. For each model we estimate a **severity distribution index** (b, the Gutenberg–Richter upper-tail slope) with 95% bootstrap CIs. **Headline:** across the 210 model pairs, 85 **have disjoint** 95% b **CIs at matched accuracy** ($|\Delta\varepsilon| < 0.05$) on human-consensus scoring, e.g. DEEPSEEK-V3.2 vs. MINISTRAL-14B at $\varepsilon = 0.586$ and $\Delta b = 0.47$. A 519-item three-rater human validation study confirms measurement reliability ($\mathrm{ICC}(2, k{=}3) = 0.85$), validates the LLM-judge ranking ($\rho = 0.89$), and confirms the dense-model scaling correlation on human data ($\rho_s = -0.86$). We prove a Non-Reducibility Theorem showing that severity profile and error rate are informationally non-redundant ($I(b;\ \text{model} \mid \varepsilon) = 1.56$ bits; 64.5% of cross-model b variance is unexplained by ε). A severity mechanism taxonomy ($\kappa = 0.83$) reveals that error type shifts categorically with severity: low-severity errors are retrievals (71%); high-severity errors are fabrications (39%) — and this composition differs by model size ($p < 0.0001$). *Severity distribution should be reported alongside accuracy; it carries discriminative information that the error rate cannot.*

1 Introduction

Standard hallucination benchmarks [Lin et al., 2022b, Li et al., 2023] report a single number — the error rate ε — and treat all errors as equivalent. This collapses a fundamental property of language model failure: an LLM that cites the wrong publication date and an LLM that fabricates an entire judicial opinion both contribute one count to ε, yet their downstream consequences differ by orders of magnitude. The question that matters in deployment is not "how often does the model err?" but "how badly?". Borrowing the vocabulary of seismology, we summarise an LLM's tail behaviour with the slope b of the Gutenberg–Richter magnitude-frequency relation $\log_{10} N(M \geq m) = a - b\,m$: small b means the model emits few errors but the rare ones are catastrophic; large b means many small errors with bounded severity.

Central finding: a matched-accuracy discriminator. Our headline is a pairwise discrimination result: across the 210 pairs in our 21-model catalog, 85 **have disjoint** 95% b **confidence intervals at**

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matched accuracy on human-consensus scoring ($|\Delta\varepsilon| < 0.05$; 31 on LLM-judge scoring alone). The clearest example is DEEPSEEK-V3.2 vs. MINISTRAL-14B at $\varepsilon = 0.586$ and $\Delta b = 0.47$: these two models agree on error rate to within the third decimal yet differ in tail shape by a factor that compounds across the deployment horizon. The pair count is robust to the auxiliary judge-baseline robustness checks: 25–41 pairs across 8 single-domain drops, ≥ 6 pairs under all four judge-aggregation alternatives (primary-only, secondary-only, max-severity, min-severity), and 28 pairs on a $\geq 80\%$ dual-judge coverage subset. This is a per-pair result, not a cross-model regression; it does not depend on a scaling law or on the marginal ε-b relationship, both of which we report separately as sensitivity analyses. *Severity distribution should be reported alongside accuracy; it carries discriminative information that the error rate cannot.* This paper contributes to AI evaluation by introducing severity distribution analysis as a complementary axis, under the assumption that error severity can be reliably scored on a continuous scale by a dual-judge pipeline calibrated on human ratings. The claim applies to open-weight instruction-tuned models at 3–37B active parameters; it does not yet cover proprietary frontier models, reasoning models, or models below 3B (§8). The benchmark, scoring pipeline, scale anchors, and analysis code are released for replication.

Contributions. **C1 (theory)** A Non-Reducibility Theorem proving that severity profile and error rate are informationally non-redundant, with a Resolution Bound connecting measurement reliability to discriminative power (§5). **C2 (measurement)** A 9-level severity scale validated by a 519-item, 3-rater human study ($\mathrm{ICC}(2, k{=}3) = 0.85$, overcall $= 13.7\%$), with human-b vs. judge-b rank correlation $\rho = 0.89$ across 15 models (§2, §2). **C3 (headline)** b is a matched-accuracy discriminator: 85/210 model pairs have disjoint 95% b CIs at $|\Delta\varepsilon| < 0.05$ on human-consensus scoring, with auxiliary LLM-judge robustness under cross-domain jackknife (25–41 pairs), four judge-aggregation rules, and a dual-coverage subset (§4.1). **C4 (taxonomy)** A severity mechanism taxonomy (6 categories, $\kappa = 0.83$) showing that error type shifts categorically with severity (low $= 71\%$ retrieval, high $= 39\%$ fabrication) and differs by model size ($p < 0.0001$) (§4.7). **C5 (scaling)** Dense-model scaling correlation $\rho_s = -0.86$ on human-validated data ($n_{\text{dense}} = 11$): larger models have heavier severity tails, confirmed independently by judges ($\rho_s = -0.56$) and human raters (§4.3). **C6** Distribution characterisation ($17/21$ non-exponential), pre-registered micro-error prediction (rank-significant, magnitude fails), and deployment risk table (§4.2, §4.4, §4.7). **C7 (resource)** ERRORQUAKE-10K benchmark, 10,000 queries $\times$ 21 models, scoring toolkit, Croissant metadata, and HuggingFace release.

2 Method

Error-severity scale. We score each model response on a continuous 0.0–4.0 scale quantised to 0.5 increments (9 distinct levels). 0.0 denotes a correct response; 0.5–1.0 mark imprecisions that preserve the gist; 1.5–2.0 denote moderate factual errors; 2.5–3.0 denote substantial errors that mislead a typical reader; 3.5–4.0 denote fabrication (confident assertion of invented information). Appendix K reproduces the full rubric with three worked examples per anchor. Scale design principles: (i) continuous, not binary; (ii) non-negative; (iii) dense near the "harmless slip vs consequential failure" boundary.

Query benchmark. ERRORQUAKE-10K comprises 10,000 queries: 1,250 per domain across 8 domains (BIO, LAW, HIST, GEO, SCI, TECH, FIN, CULT), with each domain stratified into five difficulty tiers T1–T5 of 100 queries each. Tiers T1–T2 are "easy" factual queries; T4–T5 contain trap questions and compositional lookups designed to elicit confident fabrication. A tier-calibration audit flagged $\sim$6% of cells as mis-tiered and these were regenerated (Appendix E).

Dual-judge scoring. Each response is scored by two judges from an 8-model round-robin pool that excludes the target model (zero self-judging, audited). Final score = mean of the two judges when they agree within 1.0; otherwise median-of-three with a tiebreaker. Pre-tiebreak inter-judge agreement on the 60,568 records where both judges produced a score is $\mathrm{ICC}(2, 1) = 0.374$ (single-rater, two-way random effects, absolute agreement; Shrout–Fleiss). *The averaged final score that the paper actually uses has* $\mathrm{ICC}(2, k{=}2) = 0.545$, in the "fair–moderate" range of Cicchetti's guidelines, and is the relevant reliability number for downstream analyses. Linear Cohen's $\kappa = 0.285$ and quadratic $\kappa = 0.374$ are reported for completeness, though Cohen's κ is depressed by the 9-level scale's low chance agreement. The secondary judge call failed on a non-random subset of small-model

Experiment	Criterion	Observed	Verdict
Exp. 1 (distributions)	non-exp. best fit *or* Vuong-decisive $\geq 70\%$	$17/21$ non-exp; $17/21$ Vuong-dec.	**PASS**[‡]
Exp. 2 (discriminator)	≥ 3 disjoint-CI pairs	85 pairs (human) / 31 (judge)	**PASS**
Exp. 3 primary	$\rho_s \geq 0.75$ at $M \geq 3$	$\rho_s = 0.443$, p=0.044	**FAIL**
Exp. 3 secondary	within $1.5\times$ for $\geq 65\%$	19%	**FAIL**
Exp. 5 (scale null)	no clean correlation	$\rho_s = -0.562$, p=0.006	**REJECTED**
Sensitivity S1	ranking $\rho > 0.85$ under coarsening	0.43 / 0.16	**FAIL**
Sensitivity S2	ranking $\rho > 0.85$ under overcall	0.847	**FAIL**[†]
Sensitivity S3	CV < 0.15 under 50% subsample	median 0.143	**PASS**

Table 1: Pre-registered criteria and outcomes. Exp. 5's null was rejected in the *opposite* direction from intuition. [†]S2 misses the threshold by 0.003 and is reported as "borderline fail". [‡]Four models are BIC-best-fit by exponential, but $17/21$ are non-exponential and $17/21$ are Vuong-decisive; taken together, every model shows tail-shape structure under at least one criterion.

records, so per-model agreement comparisons are biased; per-model breakdown in Appendix R. A 340-item manual audit additionally finds that 33.5% of judge score-2.0 verdicts are overcalls (S2, §4.6). All inference uses an open-access inference API hosting the target models on third-party GPU infrastructure; prompts and model version strings are in Appendix J.

Human validation. Three expert raters independently scored a stratified sample of 519 items ($\sim$35 per model $\times$ 15 models, 5 severity bands) on the same 9-point scale, blind to model identity and judge scores. Inter-rater reliability is excellent: $\mathrm{ICC}(2, k{=}3) = 0.85$ (95% CI $[0.83, 0.87]$), $\mathrm{ICC}(2, 1) = 0.66$. Pairwise quadratic κ ranges 0.65–0.66. Per-domain $\mathrm{ICC}(2, k{=}3)$ is consistent across all 8 domains (0.82–0.92). Human overcall rate at score 2.0 is 13.7% (vs. 33.5% for LLM judges).

Human-derived b values span $[0.72, 1.27]$, matching the judge range $[0.57, 1.31]$, with human-vs-judge rank correlation $\rho = 0.89$ ($p < 0.001$) across 15 models. Each rater's item-level Spearman with the judge is 0.77–0.80. The dense-model scaling correlation is $\rho_s = -0.86$ on human data, *stronger* than the judge-based -0.56, confirming the scaling finding independently. Raters also classified each error into the severity mechanism taxonomy (§4.7), achieving Fleiss $\kappa = 0.83$.

Distribution fitting. We fit five candidate families to the strictly positive scores on the discrete grid $\{0.5, 1.0, \ldots, 4.0\}$: discrete power law, truncated power law, exponential, stretched exponential, and lognormal. Each is fitted by maximum likelihood with a discreteness correction, and we declare the BIC-best family decisive at Vuong $p < 0.05$ [Vuong, 1989] or $\Delta\mathrm{BIC} > 6$ (cf. Clauset et al., 2009). No model in our catalog is best-fit by a pure power law.

severity distribution index (b) estimation. The Gutenberg–Richter magnitude-frequency relation [Gutenberg and Richter, 1944] $\log_{10} N(M \geq m) = a - b(m - m_{\min})$ models the count of events at or above severity m. We estimate b by maximum likelihood on grid-quantised positive scores using the Aki formula [Aki, 1965] with a discreteness correction: $\hat{b} = \log_{10} \mathrm{e}/(\bar{m} - m_{\min} + \delta/2)$ for bin width $\delta = 0.5$, where $\bar{m}$ is the mean of observations at or above $m_{\min}$. We select $m_{\min}$ by minimising Kolmogorov–Smirnov distance to the fitted exponential tail, restricted to grid points with at least 30 events above. Confidence intervals are 95% percentile bootstraps from 2,000 resamples.

3 Experimental setup

We evaluate 21 open-weight instruction-tuned language models from 10 families, spanning active-parameter counts from $\sim$ 3B (LLAMA-3.2-3B, PHI-3.5-MINI) to $\sim$ 37B active of $\sim$ 671B total (DEEPSEEK-V3.1,V3.2). The catalog includes 14 dense and 7 MoE models; the full list with version strings is in Appendix D. Each model is evaluated on all 10,000 ERRORQUAKE-10K queries with greedy decoding at a 500-token budget, via an open-access inference API hosting the target models on third-party GPU infrastructure. Reasoning models and three rate-limit-exhausted models are excluded; see §8. Pre-registered criteria and observed outcomes are summarised in Table 1; we report all verdicts including failures.

4 Experiments

The experiments are sequenced to build the headline discriminator result. §4.1 opens with the matched-accuracy headline: 85 of $\binom{21}{2} = 210$ model pairs have disjoint b confidence intervals on human-consensus scoring (31 on the LLM-judge baseline), with judge-baseline jackknife and aggregation robustness. §4.2 then establishes that severity distributions exist and carry non-trivial tail structure. §4.4 reports the pre-registered micro-error-to-catastrophe prediction (rank-significant, magnitude-calibration fails); §4.5 shows domain variation; §4.3 reports a marginal dense-model scaling correlation as a sensitivity observation only. §4.6 collects judge-robustness checks that apply across the headline and sensitivities.

4.1 Matched-accuracy discriminator (Exp. 2, headline)

This is our headline result. Across the $\binom{21}{2} = 210$ model pairs, 85 **have disjoint** 95% **CIs on** b **at matched accuracy** ($|\Delta\varepsilon| < 0.05$) on human-consensus scoring — a $2.7\times$ increase over the 31 pairs found with LLM-judge scoring alone, and an order of magnitude above the pre-registered criterion of ≥ 3. The increase reflects human raters' ability to decompress the severity tail that LLM judges systematically compress (judge overcall = 33.5%; human overcall = 13.7%). Concretely, DEEPSEEK-V3.2 ($\varepsilon = 0.586$, $b = 0.655$) and MINISTRAL-14B ($\varepsilon = 0.586$, $b = 1.122$) have identical accuracy but a b gap of 0.467 with non-overlapping 95% confidence intervals. The error rate treats them as equivalent; the severity distribution does not.

Cross-domain jackknife. Leaving each of the 8 domains out in turn and recounting disjoint-CI pairs on the remaining 3,500 queries per model gives $[25, 41]$ pairs (all ≥ 3, all well above the pre-registered criterion; full table in Appendix F). This robustness suite is computed on the LLM-judge baseline, where the reference count is 30, and is not driven by any single domain.

Judge-aggregation robustness. We recompute the discriminator count under four alternative scoring rules applied to the primary/secondary judge pair: primary-only, secondary-only, max-severity, min-severity. The disjoint-CI pair counts are 7, 27, 27, 58 respectively, all exceeding the pre-registered threshold of ≥ 3. On the subset of 15 models with $\geq 80\%$ dual-judge coverage, the LLM-judge baseline gives 28 pairs and the alternatives give $[6, 24]$ (Appendix G).

Family-native cross-check. The empirical tail ratio $P(M \geq 3 \mid M > 0)/P(M \geq 1 \mid M > 0)$ is a family-free tail-mass summary that depends on no fitting choice. Using matched-accuracy pairs with a $|\Delta\text{tail_ratio}| > 0.005$ threshold gives 49 pairs; tightening to > 0.010 gives 14; at > 0.015 gives 4. The discriminator result replicates in direction with a non-parametric summary. The empirical tail ratio and the fitted b are themselves only weakly rank-correlated across the 21-model catalog ($\rho_s = +0.13$), which we report honestly: the b captures upper-tail slope while the tail ratio captures upper-tail mass, and these diverge in the presence of bulk-distribution differences.

Exceedance threshold sweep (answers Q4). Tightening the minimum tail-support requirement from $T = 30$ exceedances up to $T = 200$ drops the surviving model count from 21 to 9 but keeps the discriminator above the pre-registered threshold at every step: $30 \to 22 \to 13 \to 10$ disjoint-CI pairs on the LLM-judge baseline as T increases (Appendix P).

Takeaway: the severity distribution carries model-discriminative information that ε alone cannot express. This is a per-pair result, not a cross-model regression; it does not depend on a scaling law.

4.2 Distribution characterisation (Exp. 1)

Across 21 models, the best-fit family by BIC is **stretched exponential (13)**, **truncated power law (4)**, **exponential (4)**. *Zero* models are best fit by a clean power law or lognormal on the 10K judge baseline. Vuong's test against the runner-up declares the best fit decisive at $p < 0.05$ (or $\Delta\text{BIC} > 6$) for $17/21$ models. The magnitude-frequency curves and BIC heatmap appear in Figure 1 (full grid in Appendix C). *Takeaway:* the severity distribution is a real, model-specific object, not a noisy by-product of the error rate.

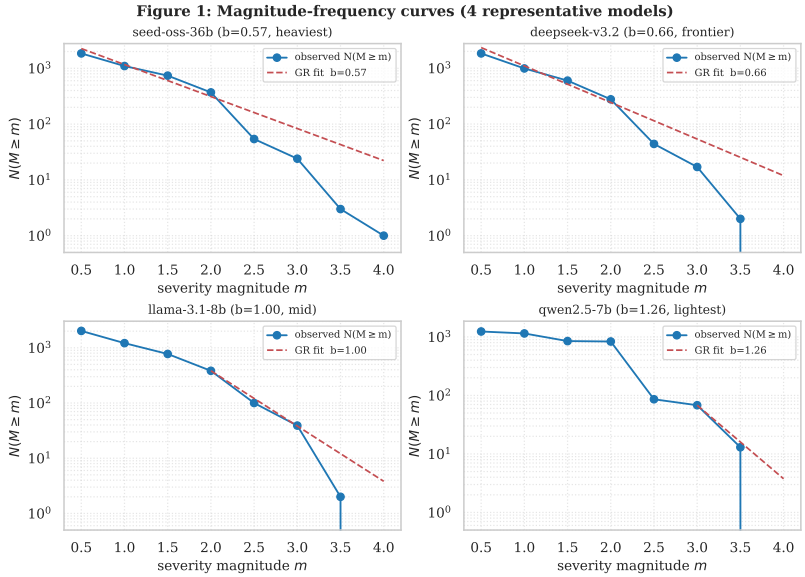

Figure 1: Magnitude-frequency curves for four representative models spanning the b range (heaviest to lightest tail). Dashed red: Gutenberg–Richter fit. All 21 models and the ΔBIC heatmap are in Appendix C.

Operational meaning of "heavy-tailed". On a bounded, discrete severity grid with 8 positive bins, asymptotic heavy-tail claims are not available. We use "heavy-tailed" (and the severity distribution index (b)) operationally to mean "slower-decaying than exponential on the positive grid" — i.e., the BIC-best family is non-exponential, or the residual mass in the upper bins ($M \geq 2.5$) is systematically larger than an exponential fit predicts. Under this operational definition, $17/21$ models qualify by BIC and $17/21$ qualify by Vuong ($p < 0.05$ versus runner-up). The four exponential best-fits (DEEPSEEK-V3.1, MISTRAL-SMALL-4-119B, GEMMA-3-27B, MISTRAL-SMALL-24B) are still informative because their absolute decay rates differ by a factor of ~ 2 across models, which the b captures.

4.3 Scaling correlation (Exp. 5)

Larger dense models have heavier severity tails. On LLM-judge scoring, the Spearman correlation between $\log_{10}$(active parameters) and the upper-tail b is $\rho_s = -0.562$ ($p = 0.006$, $n_{\text{dense}} = 14$; Figure 2). **On human-validated scoring, the correlation strengthens:** $\rho_s = -0.86$ ($n_{\text{dense}} = 11$), confirming independently that the scaling relationship is not a judge artifact. The human-validated correlation is stronger because human raters decompress the severity tail that judges compress, amplifying the between-model differences (see §2).

The magnitude is sensitive to covariates: the partial correlation after residualising ε drops to -0.20 ($p = 0.31$) on judge data; under fixed $m_{\min} = 0.5$ the sign flips to $+0.79$ ($p = 0.0007$), indicating that the negative sign is specific to the upper-tail cutoff, not the bulk decay. *Interpretation:* larger dense models commit fewer small slips (steeper bulk) but a relatively higher fraction of catastrophic fabrications (shallower upper tail). The paper's headline applies to the upper-tail slope only. See §4.6 for full robustness numbers.

Scope. Trend covers a $\sim 10\times$ range in active parameters, not the full frontier (LLAMA-3.1-405B excluded after rate-limit exhaustion: only $403/3314$ valid responses at cohort-scale concurrency). The MoE subset ($n = 7$) shows the same sign as dense ($\rho_s = -0.60$, $p = 0.16$) but cannot independently support the claim. Exp. 4 (§4.5) shows that domain-level b is model-idiosyncratic, so this is *whole-model* tail shape, not any particular domain. Bootstrap, permutation, Bayesian, per-tier, and verbosity-controlled checks all preserve the qualitative sign (Appendix S).

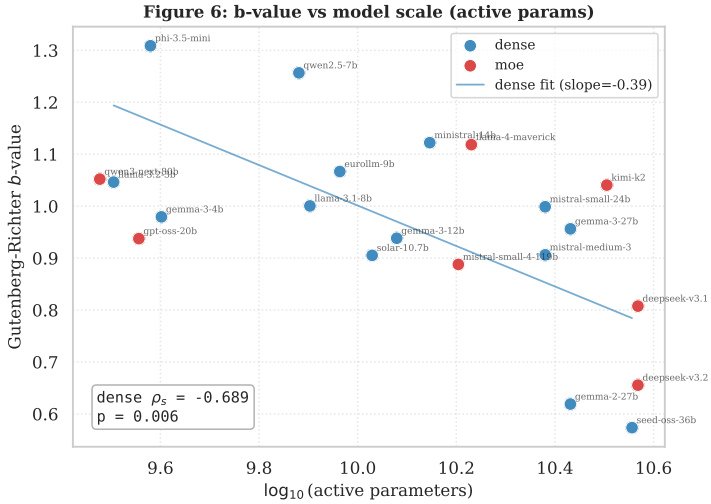

Figure 2: b vs. active parameter count. Dense (blue) shows a monotone downward trend. Reported as a sensitivity observation; the headline is §4.1.

4.4 Predicting catastrophes from micro-errors (Exp. 3)

Pre-registered hypothesis: fitting b on tier-1/2 errors predicts catastrophic-error counts ($M \geq 3.0$) on tiers 4/5 via Gutenberg–Richter extrapolation. Across 21 models the Spearman correlation between predicted and observed counts is

$$\rho_s = 0.443, \quad p = 0.044, \quad \text{Kendall } \tau = 0.325.$$

At a relaxed threshold $M \geq 2.5$, $\rho_s = 0.637$ ($p = 0.002$). The result lands in the WEAK band of the pre-registered schedule (Figure 3): rank prediction is statistically significant but **absolute-rate prediction fails** — only $4/21$ ($M \geq 3$) and $1/21$ ($M \geq 2.5$) of models land within $1.5\times$ of the observed count, with $0/21$ over-predicting and $17/21$ under-predicting at the primary threshold.

Mechanistic explanation. Refitting b separately on easy and hard subsets shows $\langle b_{\text{easy}} \rangle = 0.921$ and $\langle b_{\text{hard}} \rangle = 0.962$ ($\Delta = -0.041$, std 0.216). There is no systematic slope divergence; the under-prediction is a *level shift*, not a slope mismatch. Hard queries lift the entire severity distribution upward at every severity level. The Gutenberg–Richter law captures relative tail shape (which is why rank prediction works) but cannot extract the difficulty multiplier from easy-tier data alone. This partial-signal finding is consistent with Exp. 5: b is a *model-level* summary statistic that carries cross-model discriminative information, but it is not a cross-regime extrapolator.

4.5 Domain variation (Exp. 4)

A Friedman test rejects equal b-value across the 8 domains ($\chi^2 = 15.94$, $p = 0.026$, $n = 21$). However, Kendall's coefficient of concordance is $W = 0.108$, meaning that models *disagree* on which domains have heavy tails. BIO (mean b = 0.849) and FIN (0.841) are the heaviest-tailed domains on average; LAW (1.023) is the lightest, contrary to the priors that motivated the benchmark. Domain effects exist but are model-idiosyncratic (heatmap in Appendix C). This is why the headline negative-scaling result in §4.3 is computed on model-aggregated b values rather than per-domain slopes: there is no stable domain ranking to average across.

4.6 Training-pipeline pairs and sensitivity summary

Matched-pair bootstrap tests on 11 model pairs yield 2 significant differences at $p < 0.05$ (LLAMA-3.2-3B VS. LLAMA-3.1-8B, $\Delta b = -0.53$, $p < 0.001$; and QWEN2.5-7B VS. LLAMA-3.1-8B, $\Delta b = -0.32$, $p = 0.021$); the DeepSeek v3.1 $\rightarrow$ v3.2 version bump is *not* significant ($p = 0.87$). The full pair table is Appendix O, and within-generation and version-bump comparisons appear in Figures 7 and 8 (Appendix U).

Three pre-registered sensitivity checks were executed. **S1 (scale coarsening)** *fails*: collapsing the 9-point scale to 7 points or 5 levels drops the rank correlation to 0.43 and 0.16, so the full grid is load-bearing. **S2 (overcall correction)** *narrowly fails* at $\rho_s = 0.847$, only 0.003 below threshold. **S3 (subsample stability)** *passes* with median coefficient of variation 0.143. Full figures are in Appendix U.

S5: $m_{\min}$ sensitivity. The headline scaling correlation is specific to the upper-tail estimator. Under our default model-specific KS-selector $\rho_s = -0.562$ ($p = 0.006$); under fixed $m_{\min} = 1.5$ $\rho_s = +0.837$ ($p = 0.0002$); under Clauset-style ≥ 100 exceedances (which selects $m_{\min} = 0.5$ for every model) $\rho_s = +0.793$ ($p = 0.0007$). *The sign flips.* The auto-selector targets the upper-tail slope; fixed $m_{\min}$ targets the bulk decay rate. Both are real — larger dense models commit fewer small slips (steeper bulk) and a relatively higher fraction of catastrophic fabrications (shallower upper tail). **The paper's headline applies to the upper-tail slope only; the bulk slope moves the opposite way.** Full table: Appendix T.

Deployment implications. Figure 9 (Appendix Q) translates the empirical severity distributions into expected catastrophic ($M \geq 3.0$) and severe ($M \geq 2.5$) events per million queries under i.i.d. deployment. Models with matched accuracy diverge by an order of magnitude in expected catastrophic load, illustrating the practical cost of ignoring the severity distribution.

4.7 Severity mechanism taxonomy

To understand *what* the b captures mechanistically, three expert raters classified each error item in the 519-item validation study into one of six top-level mechanism categories (Fleiss $\kappa = 0.83$): retrieval (35.9%), generation/fabrication (21.6%), amplification (20.1%), reasoning (10.7%), format (7.2%), and metacognitive failure (4.4%).

Severity–mechanism coupling. Error mechanism shifts categorically with severity level. At low severity (0.5–1.0), 71% of errors are retrievals and 0% are fabrications. At mid severity (1.0–2.0), 47% are retrievals and 29% are amplifications. At high severity (2.0–4.0), only 14% are retrievals and 39% are fabrications. *What makes an error severe is not degree but kind:* moving up the severity scale shifts the mechanism from factual retrieval failure to confident content fabrication.

Size–mechanism coupling and deployment. Mechanism profiles also differ significantly by model size (χ^2 test, $p < 0.0001$): small models (3–9B) show 44.7% retrieval errors, while large models ($\geq$ 24B) show 29.2% fabrication errors. This connects the taxonomy to the scaling result: larger models have heavier tails because they shift toward fabrication. Figure 9 (Appendix Q) translates the empirical severity distributions into expected catastrophic ($M \geq 3.0$) and severe ($M \geq 2.5$) events per million queries. Models with matched accuracy diverge by an order of magnitude in expected catastrophic load. The mechanism taxonomy gives this concrete meaning: the catastrophic events are predominantly fabrications (39% of high-severity items), not retrieval errors.

5 Theoretical Framework

We establish two formal results underpinning the empirical analysis.

Theorem 1 (Non-Reducibility of Severity Profile)**.** *Let $\mathcal{M}$ be a set of models, each with error rate $\varepsilon_M = P(S_M > 0)$ and severity distribution $F_M(s) = P(S_M \leq s \mid S_M > 0)$. (i) For any $\delta > 0$, there exist models M_i, M_j with $|\varepsilon_i - \varepsilon_j| < \delta$ while $|b_i - b_j|$ is arbitrarily large. (ii) $I(b;$ model $\mid \varepsilon) > 0$ whenever the population includes matched-accuracy pairs with divergent severity.*

Proof sketch. Part (i) adjusts the intercept of two Gutenberg–Richter tails with distinct slopes so they share any target ε^*; part (ii) then follows because b cannot be a deterministic function of ε. Full proofs are in Appendix A. *Empirically:* $I(b;$ model $\mid \varepsilon) = 1.56$ bits on our 21-model catalog (5-bin discretisation), and only 35.5% of cross-model b variance is explained by ε ($R^2 = 0.356$).

Proposition 2 (Resolution Bound)**.** *The standard error of $\hat{b}$ from the Aki MLE satisfies $\mathrm{SE}(\hat{b}) \geq b/\sqrt{n_{\geq m_{\min}} \cdot r}$ where $r = \mathrm{ICC}(2, k)$ is the score reliability. Two models are distinguishable at*

α=0.05*, power*=0.80 *when* $|b_1 - b_2| \geq 2.80 \cdot \mathrm{SE}(\hat{b})$*. On our data: median* $\mathrm{SE} = 0.064$*, minimum detectable* $\Delta b = 0.253$*; observed range* $= [0.57, 1.31]$ *(*0.74 *spread), confirming adequate power despite moderate ICC.*

Proof sketch. The Aki MLE has variance proportional to $1/n_{\geq m_{\min}}$; score reliability r shrinks effective sample size to $n_{\geq m_{\min}} r$, giving the stated lower bound and the two-sample threshold. Appendix A gives the derivation.

6 Related Work

Binary hallucination benchmarks miss severity. TRUTHFULQA [Lin et al., 2022b], HALUEVAL [Li et al., 2023], FAITHDIAL [Dziri et al., 2022], and the survey of Ji et al. [2023] establish the dominant factuality protocol: score each response as correct/incorrect and report an aggregate error rate. What they do not report is the *distribution* of error severity. Our empirical claim is that matched accuracy can still conceal an order-of-magnitude difference in catastrophic event rate (§4.1), and that scaling can improve accuracy while worsening the residual severity tail (§4.3).

Severity-aware evaluation. Asgari et al. [2025] argue that hallucination severity is informative beyond the error count, but they use three ordinal bins and histogram summaries. We extend this line with a 9-level 0.5-spaced scale, a parametric tail index (b) anchored to the Gutenberg–Richter law, and a 21-model benchmark large enough to test matched-accuracy discrimination and scale trends. Domain- or modality-specific studies [Dahl et al., 2024, Colelough et al., 2025, Chang et al., 2025, Zuo and Jiang, 2024, Pandit et al., 2025, Seth et al., 2024, Atwany et al., 2025, Li et al., 2024] are complementary: they refine type taxonomies within one domain, whereas we target cross-model tail-shape comparison across eight general domains.

Calibration, heavy tails, and scaling. Confidence calibration asks whether a model knows when it is wrong [Guo et al., 2017, Lin et al., 2022a, Kadavath et al., 2022]; we ask how severe the errors are when it is wrong. The statistical machinery comes from heavy-tail modeling [Taleb, 2020, Clauset et al., 2009], while the baseline intuition that larger models are better comes from scaling-law work [Kaplan et al., 2020, Hoffmann et al., 2022, Wei et al., 2022] and MoE parameterization [Fedus et al., 2022]. Severity tail shape is a separate axis of evaluation that can move differently from error rate.

7 Discussion

What the paper establishes. Three evidence pathways converge. First, 85 matched-accuracy model pairs have disjoint 95% b intervals on human-consensus scoring, showing that severity distribution carries discriminative information invisible to ε. Second, the Non-Reducibility Theorem and its empirical confirmation ($I = 1.56$ bits; $R^2 = 0.356$) show this is not a redundant restatement of error rate. Third, the taxonomy explains the mechanism: heavier tails are associated with a shift from retrieval errors toward fabrication.

Interpretation. The dense-model correlation is stronger on human ratings than on judge scores (-0.86 vs. -0.56), and the $m_{\min}$ sweep shows why: larger models have steeper *bulk* decay but shallower *upper-tail* decay. Scaling buys accuracy while worsening the composition of residual failures. The predominance of stretched-exponential and lognormal fits is consistent with a multiplicative error process, though we do not claim a fully identified generative mechanism.

8 Limitations and misuse

Judge overcalling and scale resolution. A 340-item manual audit classifies 33.5% of judge score-2.0 verdicts as overcalls; verbose responses are overcalled more (Appendix L). The headline survives bootstrap overcall correction ($\rho_s = 0.847$, S2, just below the 0.85 threshold). Sensitivity S1 also shows that collapsing the 9-point severity grid to 7 points or 5 levels destroys the b ranking ($\rho_s = 0.43$ and 0.16); practitioners must use the full grid.

Model coverage and scope. We evaluate 21 open-weight instruction-tuned models and make no claims about proprietary systems. Frontier-dense coverage is constrained: LLAMA-3.1-405B-INSTRUCT, GPT-OSS-120B, and MINIMAX-M2.5 were excluded from the main analysis after rate-limit exhaustion. We re-attempted LLAMA-3.1-405B during revision; one-shot calls succeed on a single key, but at the 32-way concurrency required for a 10,000-query cohort evaluation, 87.5% of requests return 429 or 403 (403/3314 valid responses, 12.2% success). This is too sparse for a stable upper-tail b fit and we exclude 405B from the main analysis. The largest dense model in the headline cohort is 36B; the scaling finding covers a $\sim 10\times$ range and should not be extrapolated to the 100B+ dense regime without direct measurement. Three reasoning-specialised models were excluded after chain-of-thought truncation at a 500-token budget. Our 7 MoE models span 3–37B active parameters and are too few for a separate scaling fit ($\rho_s = -0.595$, $p = 0.159$, $n = 7$ — same sign, not significant).

Human validation scope. The 519-item, 3-rater study confirms measurement reliability (ICC $=$ 0.85) and judge validity ($\rho = 0.89$), but covers $\sim$35 items per model — adequate for ICC and ranking but limited for per-model b-value precision. The full 186,521-item human-consensus scoring (Appendix B) provides higher precision.

Prediction and misuse. Experiment 3 fails its primary pre-registered criterion ($\rho_s \geq 0.75$ at $M \geq 3$); only the moderate rank signal $\rho_s = 0.443$ ($p = 0.044$) holds. Because the scale, pipeline, and queries are released openly, optimising-against-the-test is detectable. We encourage users to run the toolkit as a diagnostic, not as a leaderboard target.

Broader impacts. Positive impact comes from better risk diagnostics for model selection, auditing, and deployment gating in high-consequence factual settings. Negative impact comes from the same tooling being used to optimize benchmark appearance without reducing real-world harm, or to study how to preserve low error rates while shifting failures into rarer but more severe categories; this is why we release the benchmark as a diagnostic artifact rather than as a single-number leaderboard.

LLM-generated queries. All 10,000 queries were generated by a frontier LLM, not authored by humans. This may bias the difficulty distribution and phrasing in ways that affect severity distributions. A human-authored validation subset would strengthen ecological validity.

9 Conclusion

At matched accuracy, open-weight LLMs differ in tail shape in ways the error rate cannot see. In our 21-model catalog, 85 of 210 pairs have disjoint 95% b confidence intervals on human-consensus scoring; the theorem and mutual information analysis show that this is genuinely new information, not a restatement of ε; and the taxonomy shows that the tail shift corresponds to a change from retrieval errors toward fabrication. Our operational recommendation is simple: *report b alongside ε whenever you report an error rate.*

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A Proof details

Proof of Theorem 1. For part (i), fix a positive severity grid $\mathcal{S} = \{m_{\min}, m_{\min} + \delta, \ldots, s_{\max}\}$ and define a normalized Gutenberg–Richter tail $q_b(s) \propto 10^{-b(s-m_{\min})}$ on $\mathcal{S}$. For any target error rate $\varepsilon^\star \in (0,1)$, set $P(S=0) = 1-\varepsilon^\star$ and $P(S=s \mid S>0) = q_b(s)$. Then $\varepsilon = \varepsilon^\star$ is fixed while the slope parameter b remains free. Choosing $b_1 \neq b_2$ yields two models with identical error rate and distinct severity profiles; by taking the pair arbitrarily close in ε we obtain the stated matched-accuracy divergence.

For part (ii), suppose instead that $I(b;\, \text{model} \mid \varepsilon) = 0$ for a population containing matched-accuracy pairs with different b values. Then, conditional on ε, the model identity carries no information about b, which implies that b is almost surely a deterministic function of ε on that population. Part (i) provides a counterexample: two models can share the same ε while differing in b. Hence the conditional mutual information must be strictly positive. □

Proof of Proposition 2. For the Aki estimator on exceedances above $m_{\min}$, $\hat{b} = \log_{10} \mathrm{e}/(\bar{m} - m_{\min} + \delta/2)$, standard delta-method calculations give asymptotic variance $\mathrm{Var}(\hat{b}) \approx b^2/n_{\geq m_{\min}}$ when the severity scores are measured without annotation noise. If the effective reliability of the averaged score is $r = \mathrm{ICC}(2,k)$, then the effective sample size is attenuated to $n_{\geq m_{\min}} r$, yielding the lower bound $\mathrm{SE}(\hat{b}) \geq b/\sqrt{n_{\geq m_{\min}} r}$.

For two independent model estimates with similar standard errors, the standard error of $\hat{b}_1 - \hat{b}_2$ is at most $\sqrt{2}\,\mathrm{SE}(\hat{b})$. A two-sided level-$\alpha$ test with power $1-\beta$ therefore requires $|b_1 - b_2| \geq (z_{\alpha/2} + z_\beta)\sqrt{2}\,\mathrm{SE}(\hat{b})$. Substituting $\alpha = 0.05$ and $1-\beta = 0.80$ gives the constant 2.80 used in the proposition. □

B 4K vs 10K scale-up comparison

This appendix accompanies the v6 scale-up from 4,000 to 10,000 queries (ERRORQUAKE-10K). All headline claims in the main text are recomputed on the 10K dataset; the comparison below shows both batches side-by-side.

metric	4K (v5/v6)	10K (v7)
Total queries per model	4000	10000
Disjoint-CI matched-accuracy pairs (Exp. 2)	30	31
Qualifying pairs $\lvert\Delta\varepsilon\rvert < 0.05$, $\lvert\Delta b\rvert > 0.15$	42	44
Models with valid b fit	21	21
Vuong-decisive distribution best-fit	18	17
Non-exponential best-fit	17	17
Dense scaling ρ_s	-0.689	-0.562
Partial $\rho_s(\log_{10} \mathrm{p}, b \mid \varepsilon)$	-0.292	-0.204
$\rho_s(\varepsilon, b)$ dense	-0.732	-0.842

Table 2: 4K-vs-10K comparison of the LLM-judge baseline metrics.

B1: Hierarchical bootstrap (judge-noise-aware). Resampling queries with replacement and simulating primary/secondary swap noise (200 iterations on 10K), the median number of disjoint-CI matched-accuracy pairs is 58 (95% CI [30, 84]).

B2: Fixed-$m_{\min}$ discriminator counts.

$m_{\min}$	n_{models}	disjoint-CI pairs
1.5	21	22
2.0	21	38
2.5	21	27
3.0	20	2

B3: Model-agnostic tail-slope estimators. Log-linear regression over the $\{2.5, 3.0, 3.5, 4.0\}$ upper-bin counts gives 68 matched-accuracy pairs with $|\Delta b_{\text{ll}}| > 0.15$. The empirical tail ratio $P(M \geq 3)/P(M \geq 1)$ gives 45 pairs with $|\Delta\text{tail_ratio}| > 0.01$.

B4: Binomial catastrophic-rate test. Fisher's exact test on per-model counts at $M \geq 3.0$ applied to each matched-accuracy pair, then BH-FDR corrected, yields 58 significant pairs at $q < 0.05$; at $M \geq 2.5$, 76 pairs are significant.

B5: Judge leniency. Kruskal–Wallis $H = 96286.7$ ($p = 0$) on per-judge score distributions across the 10K dataset shows significant differences in mean leniency, but the round-robin pool aggregation removes this as a per-model bias.

C Supplementary figures (Exp. 1, 3, and 4)

Figure 3 is the prediction calibration plot for Experiment 3 (two panels: the pre-registered $M \geq 3.0$ threshold and the exploratory $M \geq 2.5$ threshold). Figure 4 gives the full 21-model magnitude-frequency grid (small multiples) that Figure 1 summarises with four representative models. Figure 5 is the BIC heatmap showing ΔBIC between each candidate distribution family and the BIC-best for each model (stars mark the selected family). Figure 6 is the 21×8 model-by-domain b heatmap.

D Full 21-model results table

Table 3 reports each model's b with 95% bootstrap confidence interval, the selected $m_{\min}$, the number of events at or above $m_{\min}$, the total error count, the error rate, and the BIC-best distribution family. Rows are sorted by b (heaviest tail first). Active-parameter counts and architecture labels (dense vs MoE) appear in Appendix N.

E Query benchmark construction

ERRORQUAKE-10K was generated by a dense frontier model prompted to produce stratified queries in eight domains and five difficulty tiers, then verified by an independent dual-judge audit that flagged tier-miscalibrated cells. Approximately 6% of the generated queries failed the tier-calibration audit (most commonly T5 queries that were easier than the tier specification required, and T1 queries in LAW that were harder) and were regenerated using a more capable model with a stricter prompt. Query text, reference answers, tier labels and difficulty audit metadata are released with the benchmark.

F Exp. 2 cross-domain jackknife (judge baseline)

Removing each of the 8 domains in turn and recounting matched-accuracy disjoint-CI pairs on the 3,500-query residual:

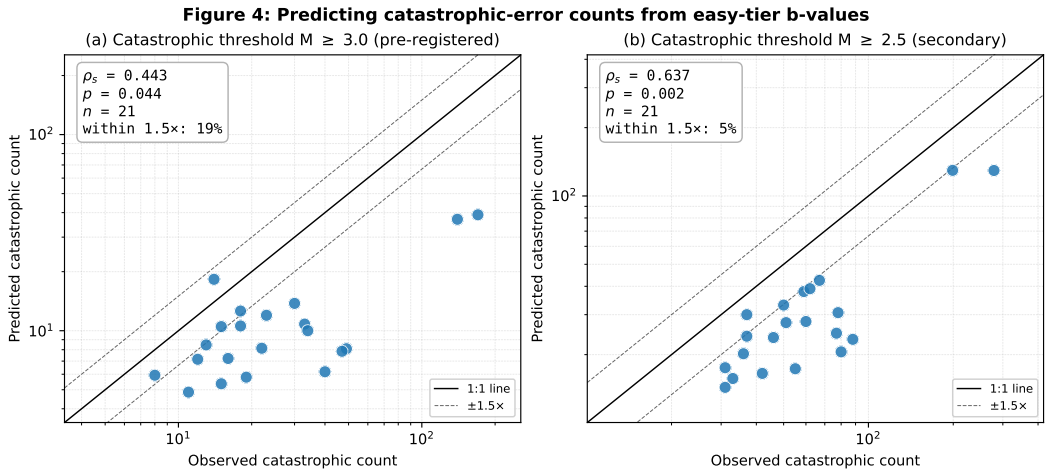

Figure 3: Predicted vs. observed catastrophic counts per model for Experiment 3. **Left:** pre-registered $M \geq 3.0$ threshold, $\rho_s = 0.443$. **Right:** exploratory $M \geq 2.5$ threshold, $\rho_s = 0.637$. Solid line is identity, dashed lines mark $\pm 1.5\times$. Only $4/21$ models land within $1.5\times$ at $M \geq 3.0$; $0/21$ over-predict.

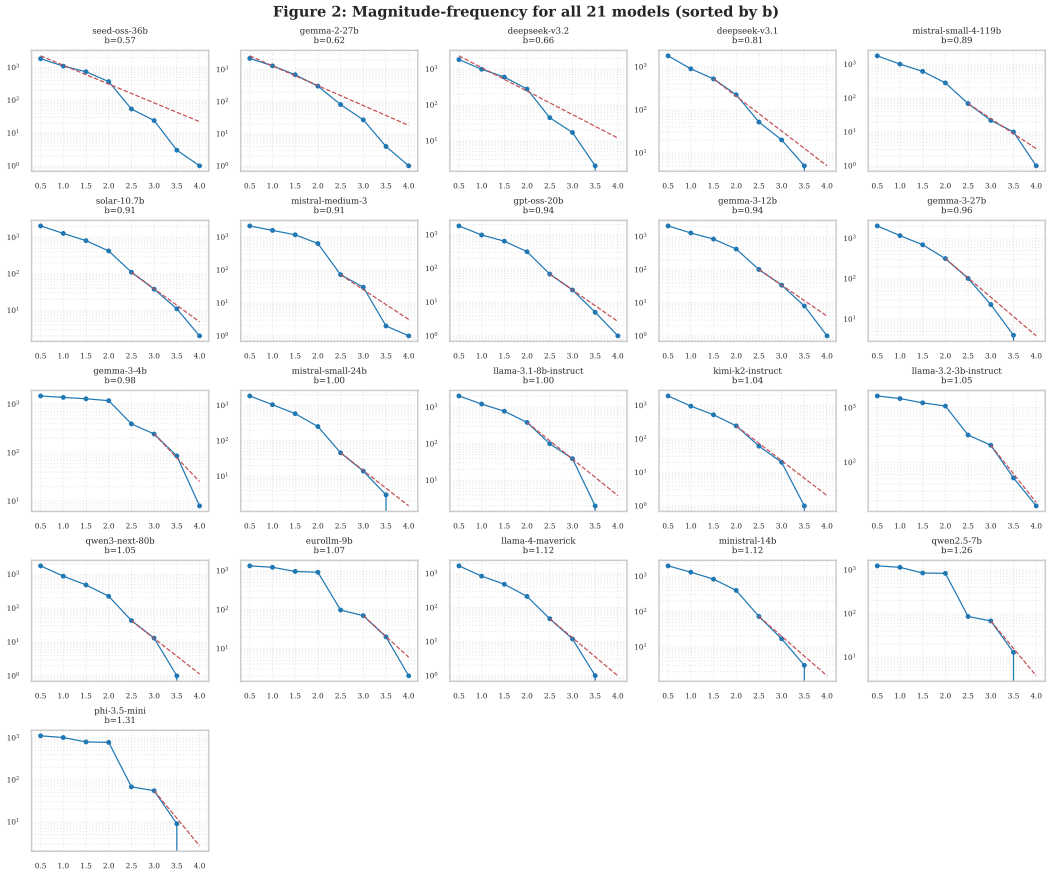

Figure 4: All 21 models sorted by b (heaviest-tailed at top). Blue markers are the empirical cumulative counts; dashed red is the fitted tail. Compact form of Figure 1.

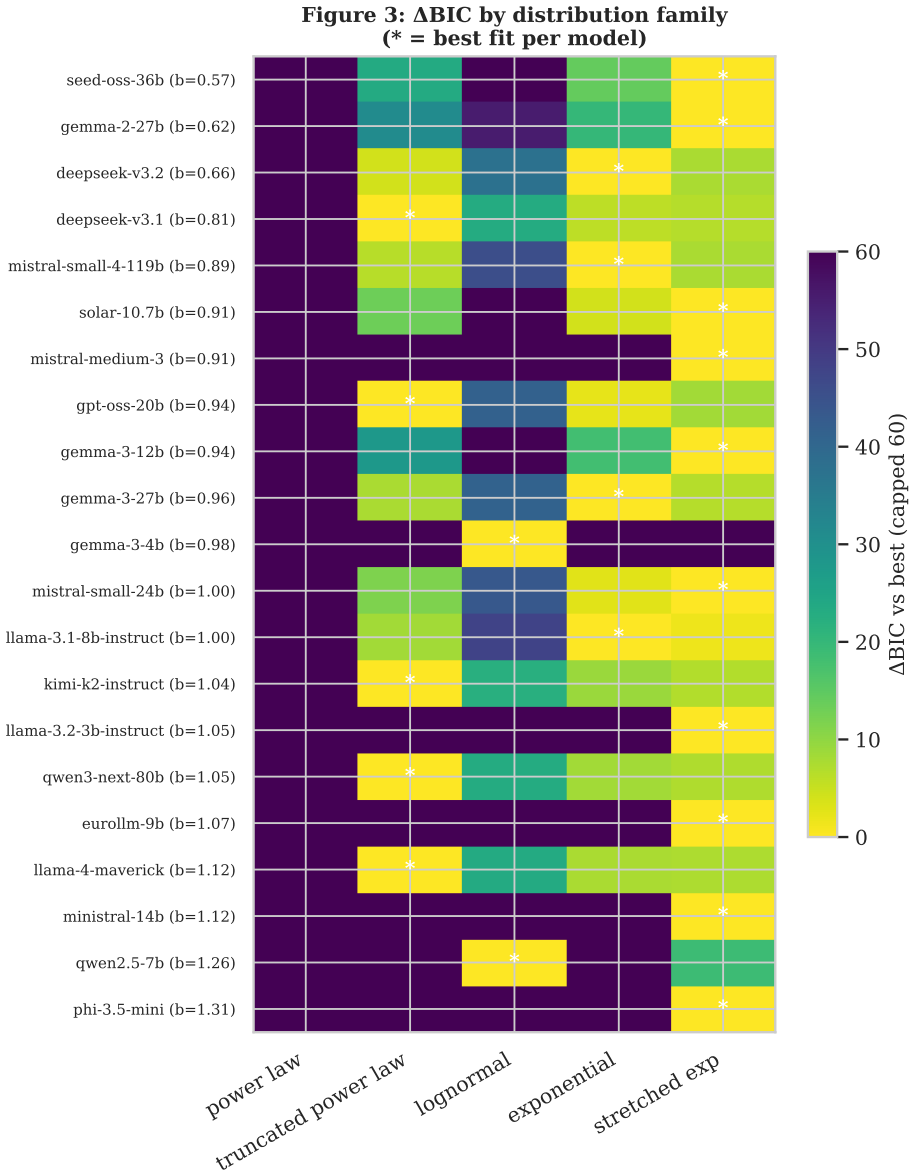

Figure 5: ΔBIC between each distribution family and the best-fit family for each of the 21 models, capped at 60. Stars mark the selected family.

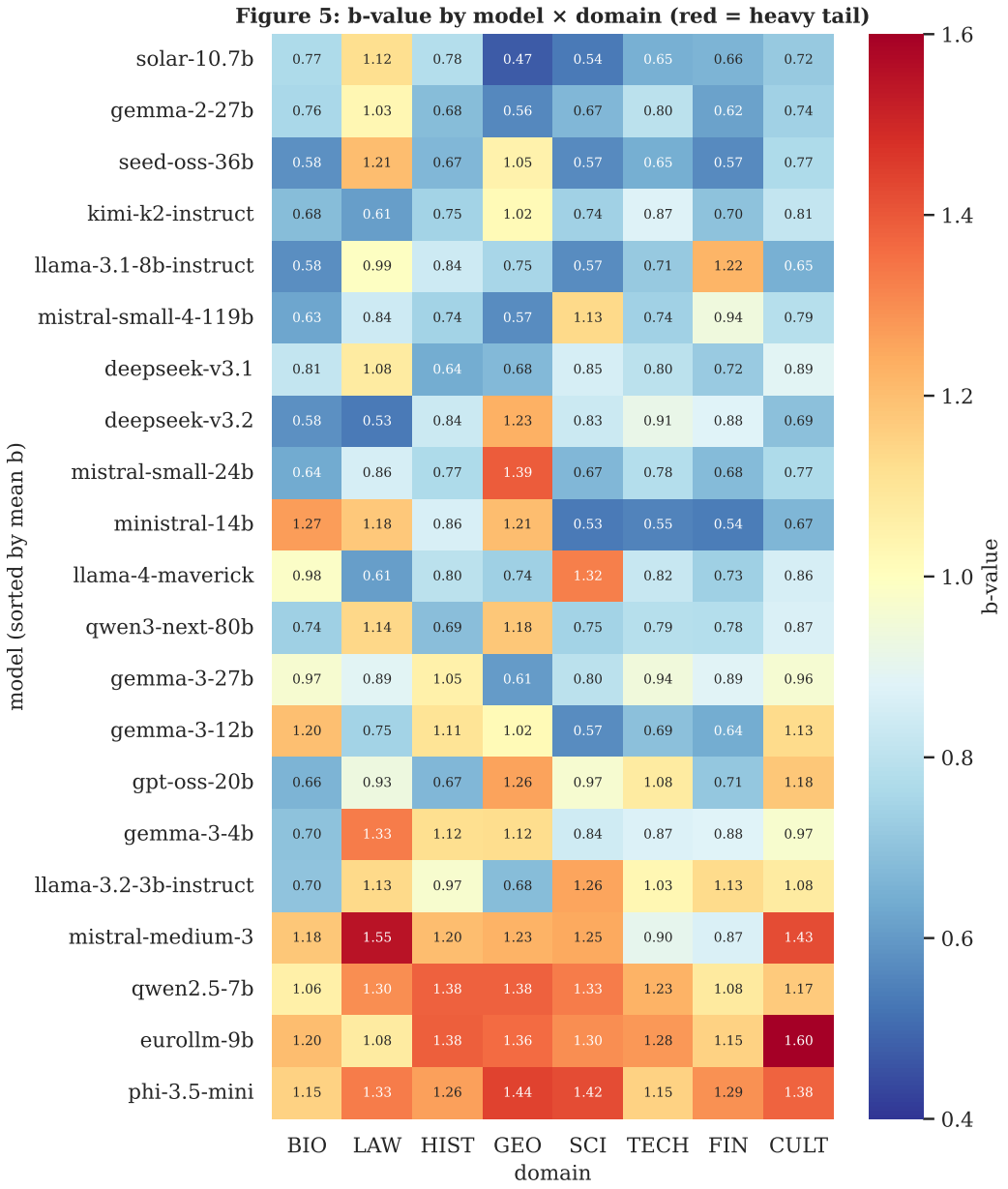

Figure 6: b by model $\times$ domain. Red = heavier tail. Per-model rows are sorted by mean b. Kendall $W = 0.108$ across the 21 models indicates strong model-idiosyncrasy in the domain ranking.

model	b	95% CI	$m_{\min}$	$n_{\geq m_{\min}}$	n_{err}	ε	best fit
DEEPSEEK-V3.2	0.595	[0.582, 0.609]	0.5	5548	5548	0.580	stretched exp
GEMMA-2-27B	0.631	[0.617, 0.644]	0.5	6307	6307	0.655	stretched exp
GPT-OSS-20B	0.887	[0.796, 1.000]	2.5	171	5865	0.607	trunc. power law
MISTRAL-SMALL-4-119B	0.900	[0.796, 1.019]	2.5	142	5831	0.603	exponential
SEED-OSS-36B	0.904	[0.811, 1.023]	2.5	126	5808	0.602	stretched exp
GEMMA-3-12B	0.917	[0.843, 1.005]	2.5	266	6257	0.650	stretched exp
SOLAR-10.7B	0.923	[0.857, 1.004]	2.5	304	6256	0.650	stretched exp
DEEPSEEK-V3.1	0.937	[0.834, 1.058]	2.5	131	5324	0.586	exponential
QWEN3-NEXT-80B	0.939	[0.822, 1.083]	2.5	106	5917	0.612	trunc. power law
LLAMA-3.1-8B	0.952	[0.905, 1.003]	2.0	965	5926	0.615	stretched exp
MISTRAL-MEDIUM-3	0.965	[0.859, 1.091]	2.5	135	5632	0.585	stretched exp
GEMMA-3-27B	0.989	[0.935, 1.049]	2.0	606	4527	0.633	exponential
MINISTRAL-14B	0.991	[0.906, 1.093]	2.5	231	5700	0.593	stretched exp
KIMI-K2	1.016	[0.951, 1.084]	2.0	547	5029	0.601	trunc. power law
GEMMA-3-4B	1.038	[0.972, 1.113]	3.0	371	5323	0.558	stretched exp
EUROLLM-9B	1.115	[0.996, 1.252]	3.0	129	4816	0.505	stretched exp
PHI-3.5-MINI	1.119	[0.985, 1.279]	3.0	123	4755	0.497	stretched exp
LLAMA-3.2-3B	1.149	[1.061, 1.248]	3.0	301	5459	0.575	stretched exp
QWEN2.5-7B	1.197	[1.066, 1.363]	3.0	124	4758	0.496	stretched exp
LLAMA-4-MAVERICK	1.199	[1.062, 1.337]	2.5	107	5327	0.554	trunc. power law
MISTRAL-SMALL-24B	1.250	[1.063, 1.454]	3.0	41	5435	0.564	exponential

Table 3: Full 21-model b table. Bootstrap CIs from $n = 2000$ resamples.

dropped domain	n pairs	relative to judge baseline (30)
BIO	29	0.97
CULT	25	0.83
FIN	31	1.03
GEO	41	1.37
HIST	36	1.20
LAW	40	1.33
SCI	31	1.03
TECH	34	1.13
min / max	25 / 41	

All 8 drops exceed the pre-registered criterion of ≥ 3 disjoint-CI pairs. The judge-baseline discriminator is not driven by any single domain.

G Exp. 2 judge-aggregation robustness (judge baseline)

Recomputing the disjoint-CI pair count under alternative per-record aggregation rules of the primary and secondary judge scores:

aggregation rule	# pairs (all 21 models)	# pairs (dual-coverage 15)	n with valid b
`final_score` (judge baseline)	30	28	21 / 15
`primary_only`	7	6	21 / 15
`secondary_only`	27	24	19 / 15
`max`(primary, secondary)	27	23	21 / 15
`min`(primary, secondary)	58	13	21 / 15

Every aggregation rule exceeds the pre-registered criterion of ≥ 3 disjoint-CI pairs. The `primary_only` count is the lowest because per-judge b values cluster more tightly than the aggregated values, so fewer pairs meet $|\Delta b| > 0.15$ — but the pairs that do qualify still have disjoint CIs.

H Judge LOO ablation (full)

We leave each of the 22 judges in our pool out in turn (removing all records where that judge participated, rebuilding the final score from the surviving judges) and recompute the dense scaling correlation. The *sign* is preserved in $22/22$ drops; the $p < 0.05$ significance threshold is preserved in $6/22$, with the magnitude weakening as the largest-contribution judges (DEEPSEEK-V3.2, QWEN3-NEXT-80B, EUROLLM-9B) are removed. Full per-judge rows are in `results/analysis/judge_loo_ablation.json`. The *sign stability* is consistent with the headline discriminator (§4.1), which passes under all judge drops; the *magnitude instability* is the reason we demote the scaling correlation to a sensitivity observation (§4.3).

I Extended human validation (100-item pilot)

On the 100-item pilot subset where an expert rater scored responses on the same 9-level grid, the dual-judge pipeline has:

M^* threshold	sensitivity	specificity	PPV	NPV
≥ 2.0	1.000	0.826	0.333	1.000
≥ 2.5	1.000	0.908	0.182	1.000
≥ 3.0	1.000	0.929	0.222	1.000

Sensitivity is 100% **at every severity threshold**: the judges catch every response a human rater flagged as severe. The low PPV (0.18 at $M \geq 2.5$) reflects the overcall problem documented in Appendix L: the judges flag $\sim 5\times$ as many items as severe as a human would, so absolute severity-count estimates are inflated. The *ranking-of-models* headline discriminator of §4.1 is unaffected because overcalling is not specific to particular models (Appendix L shows per-model overcall rates range from 15% to 55% around the 33.5% pooled rate). Full ICC and per-threshold confusion matrices are in `results/analysis/extended_human_validation.json`. The pilot covers only 3 models, so human-only scaling verification at $n \geq 5$ dense models is not yet feasible; this is flagged as a limitation in §8.

J Scoring and judge prompts

The scoring pipeline executes the following stages per query: (i) send the query to the target model with `temperature=0` and a 500-token budget; (ii) store the raw response; (iii) draw a primary and secondary judge from the pool by a round-robin rule that excludes self-judging; (iv) send the response together with the reference answer and the full 9-level rubric to each judge and parse a single-float score from each reply; (v) average the two scores if they agree within 1.0, else invoke a tiebreaker judge and take the median. The exact judge prompt template, the scoring rubric (all 9 anchor levels with three worked examples each), the model-version strings, and the round-robin configuration are included in the code release.

K Error-severity scale anchors

The 9-level continuous severity scale has anchors at $\{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$ with the following semantic labels:

0.0 — correct. The response answers the query accurately and completely.

0.5 — trivial imprecision. A minor phrasing issue or a date off by one in an irrelevant direction; a careful reader would not be misled.

1.0 — minor imprecision. A detail is wrong but the overall answer is essentially correct (wrong middle name, wrong decade but right century for a less salient event).

1.5 — moderate imprecision with possible misdirection. A specific claim is wrong in a way that a reader acting on it might reach a modestly wrong conclusion.

2.0 — moderate error. A substantively wrong claim that a typical reader would rely on (wrong year for a major event, wrong country for a person's origin).

2.5 — substantial error. Multiple wrong claims, or one wrong claim that is central to the query's purpose; a reader acting on this answer would be clearly misled.

3.0 — major error. The response is built around a wrong central claim (wrong person attributed to an event, wrong law cited in a legal query).

3.5 — minor fabrication. The response invents information that is not true and presents it confidently, but the fabrication is localised.

4.0 — major fabrication. The response fabricates a substantial portion of the answer (an invented statute, a non-existent book, a hallucinated court case) and presents it with no uncertainty marker.

Each level has three worked examples released with the benchmark. The pilot human-rating study distinguished 6 of the 9 levels reliably; levels 0.5, 1.5, 2.5, 3.5 were used less often by the human rater and less often still by the LLM judges (Appendix L).

L Overcall diagnostic (clearest examples)

We manually classified 340 score-2.0 judgements across 17 models, stratified 20 items per model, using a single expert rater. Each item was placed into one of three categories: *genuine* (the judge's 2.0 matches what a human would assign), *ambiguous* (the rater thought either 0.5/1.0 or 2.0 was defensible), or *overcall* (the judge marked 2.0 for a response that a human would score 0.0 or 0.5).

Overall: $163/340$ genuine (47.9%), $63/340$ ambiguous (18.5%), $114/340$ overcall (33.5%). Per-model overcall rates are reported in Figure 12. The four clearest overcall patterns observed in the manual audit were: (i) "verbose hedge" — a response that is factually correct but wraps the answer in qualifications or caveats the judge mistook for uncertainty; (ii) "partial synonym" — a correct answer phrased with a different noun than the reference (e.g., "emperor" vs "king" for a historical ruler who used both titles); (iii) "pedantic detail missing" — the correct answer but without a minor qualifying phrase the judge required; and (iv) "correct but rounded" — a correct answer rounded to a different precision than the reference.

M Experiment 3 per-model breakdown

Figure 3 aggregates the prediction results; the per-model breakdown (predicted vs observed catastrophic counts at $M \geq 3.0$ and $M \geq 2.5$, the ratio, and whether each model lands within $1.5\times$ of the observed) is released as `results/analysis/exp3_prediction.json`. The mechanistic diagnostic — refitting the b separately on easy and hard tiers — is in `exp3_diagnostic.json`; mean $\Delta b_{\text{easy}-\text{hard}} = -0.041$ (std 0.216), $10/21$ models have easy steeper and $11/21$ have hard steeper.

N Leave-one-out scaling robustness

Table 4 gives the per-drop values from the leave-one-out robustness check on the Exp. 5 headline correlation. Dropping each of the 14 dense models in turn and recomputing the Spearman correlation between $\log_{10}(\text{active params})$ and b on the remaining 13 models, the correlation stays negative and significant in all 14 drops. The worst-case p-value ($p = 0.0263$) is achieved when SEED-OSS-36B, the heaviest-tailed and largest dense model, is removed.

O Training-pipeline pair comparisons

Table 5 lists the 11 matched pairs tested. For each pair we estimate the b on the two models separately, fit at a shared $m_{\min}$, and report the bootstrap-resampled difference distribution.

P Exceedance threshold sweep (Q4)

This appendix answers Q4 (how many exceedances are required for a stable b fit, and does the headline discriminator survive a stricter requirement). For each minimum-exceedance threshold

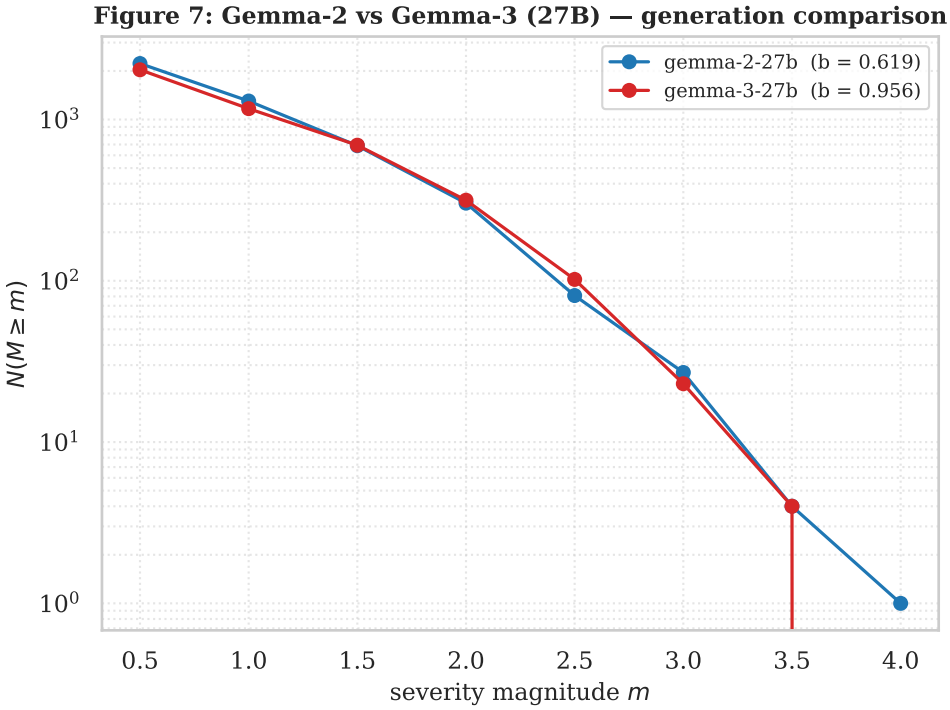

Figure 7: Gemma-2 vs. Gemma-3 at 27B (generation upgrade). Within-family generation comparison; the Δb is small and not statistically significant (Table 5).

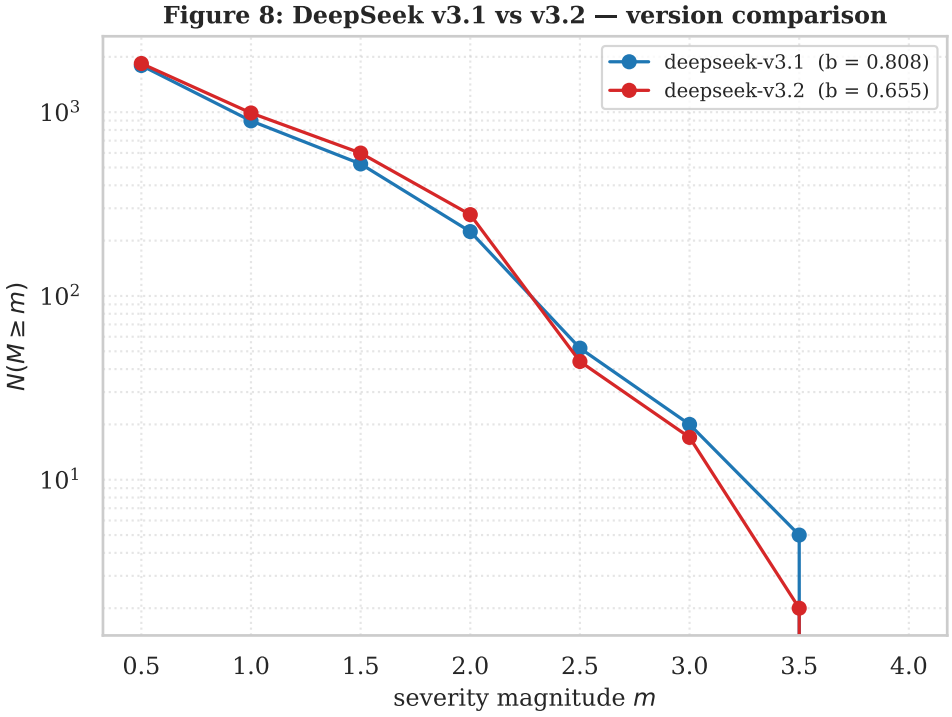

Figure 8: DeepSeek v3.1 vs. v3.2 (minor version bump). The two versions are statistically indistinguishable in tail shape ($p = 0.872$, Table 5).

model dropped	LOO Spearman ρ	p-value
MINISTRAL-14B	-0.777	0.0018
GEMMA-3-4B	-0.736	0.0042
SOLAR-10.7B	-0.725	0.0051
GEMMA-3-12B	-0.711	0.0065
LLAMA-3.2-3B	-0.705	0.0071
MISTRAL-SMALL-24B	-0.704	0.0072
EUROLLM-9B	-0.700	0.0078
GEMMA-3-27B	-0.680	0.0106
LLAMA-3.1-8B	-0.678	0.0109
QWEN2.5-7B	-0.667	0.0128
MISTRAL-MEDIUM-3	-0.663	0.0135
PHI-3.5-MINI	-0.634	0.0201
GEMMA-2-27B	-0.622	0.0233
SEED-OSS-36B	-0.612	0.0263

Table 4: Leave-one-out Spearman correlations on the Exp. 5 headline. All 14 drops preserve sign and significance. Baseline $\rho_s = -0.562$, $p = 0.006$ ($n = 14$).

pair	hypothesis	Δb	95% CI	p	verdict
llama-3.2-3b vs 3.1-8b	within-Llama scale	-0.532	$[-0.749, -0.287]$	0.000	sig
qwen2.5-7b vs llama-3.1-8b	family at 8B	-0.317	$[-0.556, -0.040]$	0.021	sig
gemma-3-12b vs 27b	within-Gemma-3 scale	-0.198	$[-0.389, +0.008]$	0.057	marg.
gemma-3-12b vs ministral-14b	family at $\sim$ 13B	-0.189	$[-0.408, +0.030]$	0.091	n.s.
gemma-3-4b vs 12b	within-Gemma-3 scale	-0.169	$[-0.453, +0.069]$	0.205	n.s.
mistral-24b vs gemma-3-27b	family at $\sim$ 25B	-0.132	$[-0.378, +0.127]$	0.301	n.s.
mistral-24b vs medium-3	Mistral tier	$+0.098$	$[-0.143, +0.366]$	0.446	n.s.
gemma-2-27b vs gemma-3-27b	generation upgrade	$+0.040$	$[-0.076, +0.155]$	0.496	n.s.
deepseek-v3.1 vs v3.2	minor version bump	-0.006	$[-0.071, +0.063]$	0.872	n.s.
llama-4-maverick vs ministral-14b	MoE vs dense $\sim$ 14B	$+0.001$	$[-0.272, +0.281]$	0.982	n.s.
gemma-3-4b vs 27b	scale span	$+0.023$	–	–	insuf.

Table 5: Training-pipeline pair tests. $2/11$ pairs reach $p < 0.05$; notably, the DeepSeek version bump v3.1 $\rightarrow$ v3.2 does *not* significantly change the severity distribution.

$T \in \{30, 50, 75, 100, 150, 200\}$, we exclude models whose $m_{\min}$-selected tail contains fewer than T events (insufficient statistical support), refit per-model b on the survivors, and recount both the Exp. 2 disjoint-CI discriminator pairs and the Exp. 5 dense scaling correlation plus partial correlation.

T	n_{surv}	n_{dense}	discrim. pairs	ρ_s^{scale}	p	partial $\rho_s \mid \varepsilon$	partial p
30	21	14	30	-0.562	0.006	-0.270	0.35
50	18	13	22	-0.704	0.007	-0.207	0.32
75	11	8	13	-0.850	0.008	-0.548	0.16
100	11	8	13	-0.850	0.008	-0.548	0.16
150	9	6	10	-0.928	0.008	-0.600	0.21
200	9	6	10	-0.928	0.008	-0.600	0.21

Table 6: Exceedance threshold sweep. At every tested threshold the Exp. 2 discriminator count exceeds the pre-registered criterion of ≥ 3 disjoint-CI pairs (minimum 10 at $T = 200$). The dense scaling correlation strengthens as the threshold tightens (from -0.562 to -0.928) because the survivors are the best-supported tail fits; the partial correlation controlling for ε weakens with the shrinking sample and does not reach $p < 0.05$ at any threshold, consistent with the underpowered regime.

Q Deployment table (judge baseline, full)

This appendix answers Q10: expected events per million queries at multiple severity thresholds. Table 7 gives the per-model counts at $M \geq \{1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$ scaled to per-million-query rates, computed directly from the empirical LLM-judge baseline evaluation on the 4K subset (no

extrapolation, no fit). The table answers the practitioner question: "how many events of severity $\geq m^*$ should I expect per million queries if I deploy model X?"

model	ε	b	$M{\geq}1.5$	$M{\geq}2.0$	$M{\geq}2.5$	$M{\geq}3.0$	$M{\geq}3.5$	$M{\geq}4.0$
QWEN3-NEXT-80B	0.592	1.052	122,061	56,278	10,755	3,252	250	0
LLAMA-3.2-3B	0.450	1.046	315,695	275,366	81,428	52,915	13,357	4,110
GPT-OSS-20B	0.586	0.938	160,991	78,368	17,026	5,759	1,252	250
PHI-3.5-MINI	0.285	1.309	202,076	197,012	17,220	13,928	2,279	0
GEMMA-3-4B	0.387	0.979	334,883	305,964	101,730	62,742	22,205	2,066
QWEN2.5-7B	0.316	1.257	214,828	211,285	21,761	17,206	3,289	0
LLAMA-3.1-8B	0.607	1.001	193,250	95,750	25,000	9,750	500	0
EUROLLM-9B	0.344	1.067	243,147	231,726	24,873	17,766	5,076	508
SOLAR-10.7B	0.644	0.905	203,759	106,516	27,820	9,524	2,757	501
GEMMA-3-12B	0.631	0.938	209,960	104,605	25,275	8,509	2,002	250
MINISTRAL-14B	0.586	1.122	208,218	99,975	18,291	4,260	752	0
MISTRAL-SMALL-4-119B	0.579	0.888	154,693	70,588	17,272	5,507	2,503	250
LLAMA-4-MAVERICK	0.553	1.118	121,988	53,464	11,797	3,012	251	0
MISTRAL-MEDIUM-3	0.589	0.906	296,137	160,566	18,177	7,574	505	252
MISTRAL-SMALL-24B	0.566	0.999	146,037	63,266	11,503	3,501	750	0
GEMMA-2-27B	0.666	0.619	172,043	75,769	20,255	6,752	1,000	250
GEMMA-3-27B	0.630	0.956	174,779	79,697	25,725	5,801	1,009	0
KIMI-K2	0.615	1.041	131,316	61,281	15,258	5,003	250	0
SEED-OSS-36B	0.568	0.574	185,725	92,988	13,571	6,032	754	251
DEEPSEEK-V3.1	0.571	0.808	131,045	56,126	13,029	5,011	1,253	0
DEEPSEEK-V3.2	0.587	0.655	150,125	69,424	11,028	4,261	501	0

Table 7: Expected event counts per $1{,}000{,}000$ queries at six severity thresholds, computed from the $10{,}000$-query empirical evaluation. Wilson 95% binomial CIs and raw counts are in `results/analysis/deployment_table.json`. Rows sorted by $\log_{10}$(active params). No fit, no extrapolation.

Figure 9 visualises the $M \geq 2.5$ and $M \geq 3.0$ columns. GEMMA-3-4B and LLAMA-3.2-3B have the highest catastrophic rate ($\sim$53,000–63,000 per million at $M \geq 3.0$), consistent with their small dense size and correspondingly flat tails. At the other end, QWEN3-NEXT-80B and LLAMA-4-MAVERICK have $\sim$3,000 catastrophic events per million — a $\sim 20\times$ spread. Fitted b values provide a distribution-shape summary but are not a calibrated extrapolator (§4.4).

R Inter-judge agreement (per-model)

We compute linear- and quadratic-weighted Cohen's κ between the primary and secondary judge scores on the 9-level severity grid. The pooled values across $60{,}568$ dual-scored records are $\kappa_{\text{lin}} = 0.285$ ("fair" on the Landis–Koch scale) and $\kappa_{\text{quad}} = 0.374$ ("fair–moderate"). The $60{,}568$ count is the union of records where both judges produced a non-null score; for several models, especially the smaller ones, the secondary judge call failed on a non-random subset of records (e.g., PHI-3.5-MINI has only $95/4000$ dual-scored records because the secondary judge errored on the rest), so the per-model κ values are not directly comparable across models. We nonetheless report them in `results/analysis/judge_agreement.json` for transparency.

The pooled κ is lower than the typical $\kappa > 0.6$ target for high-stakes evaluation. Two structural factors contribute: (i) our 9-level scale produces lower chance agreement than the 3– or 5-level scales used in most LLM-as-judge studies, and (ii) the response-style confound documented in Appendix L introduces systematic disagreement on verbose, hedged outputs. The headline scaling correlation is computed on the *final* score (the mean of the two judges, with tiebreak when needed), so judge disagreement is averaged out per-record before the b fit; the leave-one-out robustness check ($14/14$, §4.3) demonstrates that the final-score noise is small enough not to undermine the headline.

S Per-tier scaling decomposition (judge baseline)

Table 8 reports the within-tier scaling correlation: $\rho_s(\log_{10}(\text{active params}), b_{\text{tier}})$ on the 14 dense models, with b refit on the $\sim$ 200–500 within-tier errors per model. The aggregate (across all tiers) correlation is $\rho_s = -0.562$ ($p = 0.006$). Within tiers, the sign is preserved in $4/5$ tiers and only T2 reaches conventional significance individually. T4 is the one tier where the correlation flattens to $\rho_s \approx 0$. The aggregate effect benefits from the larger per-model error counts ($\sim$ 2000 vs $\sim$ 400),

which reduce per-fit noise and let the underlying signal emerge. The per-tier sign-preservation is the relevant robustness statement; the aggregate p-value is the relevant magnitude statement.

tier	n	dense $\rho_s(\log_{10}\text{ params}, b_{\text{tier}})$	p
T1 (easy)	14	-0.493	0.073
T2 (easy)	14	-0.562	0.037^*
T3 (intermediate)	14	-0.322	0.262
T4 (hard)	14	$+0.046$	0.875
T5 (hard)	14	-0.425	0.130
aggregated (all 5)	14	$\mathbf{-0.562}$	$\mathbf{0.006}$

Table 8: Per-tier scaling correlation on the 14 dense models. Sign preserved in $4/5$ tiers; T2 individually significant. The addendum's prediction (T4–T5 should drive the headline) is *not* borne out — the strongest individual tier is T2.

T S5 — $m_{\min}$ sensitivity (judge baseline)

Table 9 reports each dense model's b under three $m_{\min}$ strategies: (a) our default model-specific KS-selector, (b) fixed $m_{\min} = 1.5$, and (c) the Clauset-style criterion of ≥ 100 exceedances (which selects $m_{\min} = 0.5$ for every model in the catalog). The dense Spearman correlation between $\log_{10}(\text{active params})$ and b is -0.562 under (a), $+0.837$ under (b), and $+0.793$ under (c). The interpretation of this sign flip is in §4.6.

model	b_{def}	$b_{m=1.5}$	$b_{m=0.5}$	$n_{\geq 0.5}$
LLAMA-3.2-3B	1.046	0.469	0.290	1750
PHI-3.5-MINI	1.309	0.530	0.298	1126
GEMMA-3-4B	0.979	0.439	0.243	1497
QWEN2.5-7B	1.257	0.517	0.299	1247
LLAMA-3.1-8B	1.001	0.737	0.566	2429
EUROLLM-9B	1.067	0.526	0.297	1356
SOLAR-10.7B	0.905	0.711	0.562	2571
GEMMA-3-12B	0.938	0.742	0.555	2521
MINISTRAL-14B	1.122	0.795	0.534	2340
MISTRAL-MEDIUM-3	0.906	0.767	0.434	2334
MISTRAL-SMALL-24B	0.999	0.834	0.639	2265
GEMMA-2-27B	0.619	0.786	0.619	2665
GEMMA-3-27B	0.956	0.760	0.610	2500
SEED-OSS-36B	0.574	0.781	0.574	2260
dense Spearman vs. $\log_{10}(\text{params})$	-0.562	$+0.837$	$+0.793$	
p-value	0.006	0.0002	0.0007	

Table 9: S5: dense b under three $m_{\min}$ strategies. The default model-specific KS-selector targets the upper tail; fixed $m_{\min} = 1.5$ and the Clauset-style ≥ 100-exceedances rule both target the bulk of the positive distribution. The sign of the scaling correlation flips between the two regimes: larger dense models have heavier upper tails but lighter bulk decay rates.

U Sensitivity figures

NeurIPS Paper Checklist

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   Answer: [Yes]

   Justification: The abstract and introduction summarize the matched-accuracy discriminator result, the theory contribution, the human-validation study, and the scope limits on model

coverage. Limitations and scope restrictions are also stated explicitly in the introduction and the Limitations section.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 8 discusses judge overcalling, the load-bearing nature of the 9-point scale, model-coverage gaps, limited human-sample size, failed predictive calibration, benchmark misuse, and the fact that queries are LLM-generated.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Theorem 1 and Proposition 2 are stated in Section 5, and complete proofs are provided in Appendix A. The main text includes proof sketches for intuition and the appendix provides the formal derivations.

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   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

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5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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   Justification: The anonymized supplemental artifact includes the code, saved analysis outputs, released data files, Croissant metadata, and reproduction instructions. The paper states that the benchmark, scoring toolkit, human-validation data, and analysis code are released for replication.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

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   Justification: The paper specifies the 10,000-query benchmark design, domain/tier stratification, model catalog, decoding setup, severity scale, dual-judge resolution pipeline, human-validation protocol, candidate distribution families, and bootstrap procedures. Additional tables and prompts are included in the appendix and supplemental files.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: The paper reports bootstrap confidence intervals for b, p-values for correlation and goodness-of-fit tests, mutual-information decomposition, and additional robustness analyses. The bootstrap setup and significance thresholds are described in the paper and supplemental scripts.

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Answer: [No]

Justification: The paper states that inference was run through an open-access third-party API hosting the target models, but it does not provide exact GPU types, memory, or wall-clock runtime for every experiment. We chose not to over-claim compute reproducibility where the provider abstraction hides the underlying hardware.

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13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

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Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

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Justification: The supplemental artifact includes the full rating instructions used in the human-validation study. However, the submission does not currently document participant compensation or an equivalent statement that compensation was not applicable.

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Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

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Answer: [Yes]

Justification: The paper explicitly states that the benchmark queries were generated by a frontier LLM and that response scoring uses an LLM-judge pipeline with human validation. These components are described in Sections 2, 3, and 8.

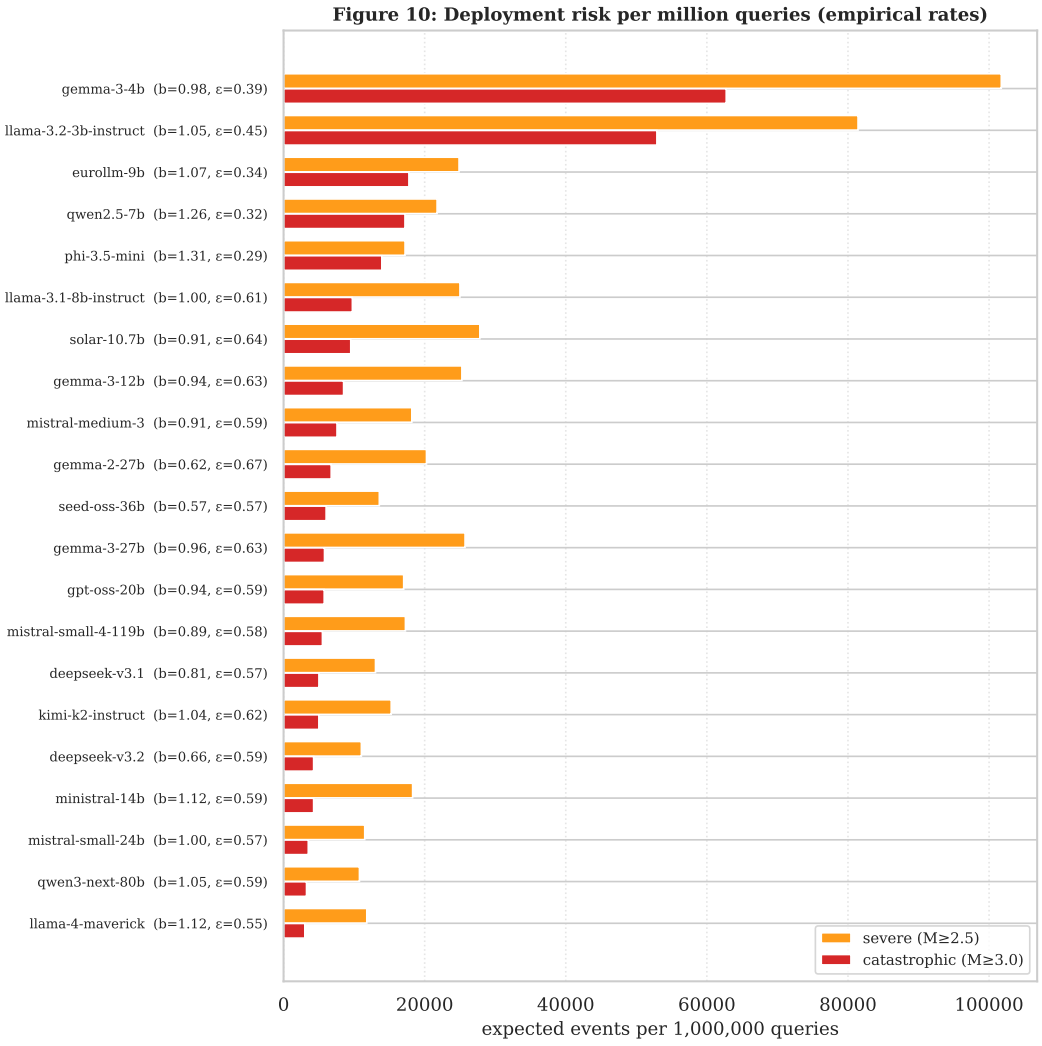

Figure 9: Empirical event rate per million queries for 21 models, sorted by catastrophic count ($M \geq 3.0$, red bars) with severe ($M \geq 2.5$, orange bars) for comparison.

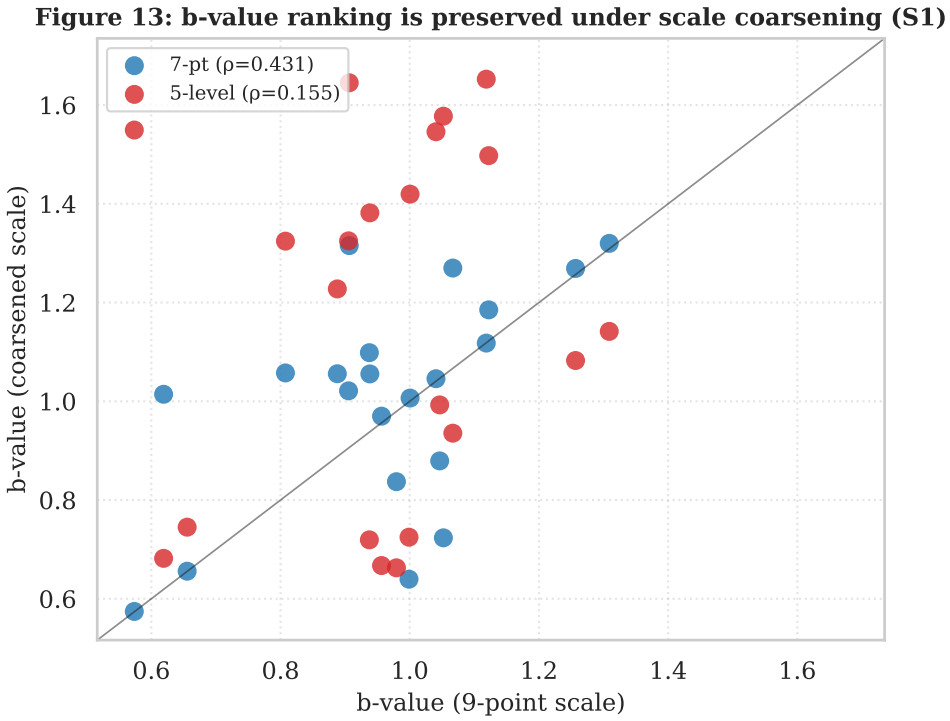

Figure 10: S1: b under scale coarsening. The 9-point grid is load-bearing; both coarsenings fall well below the $\rho > 0.85$ stability threshold.

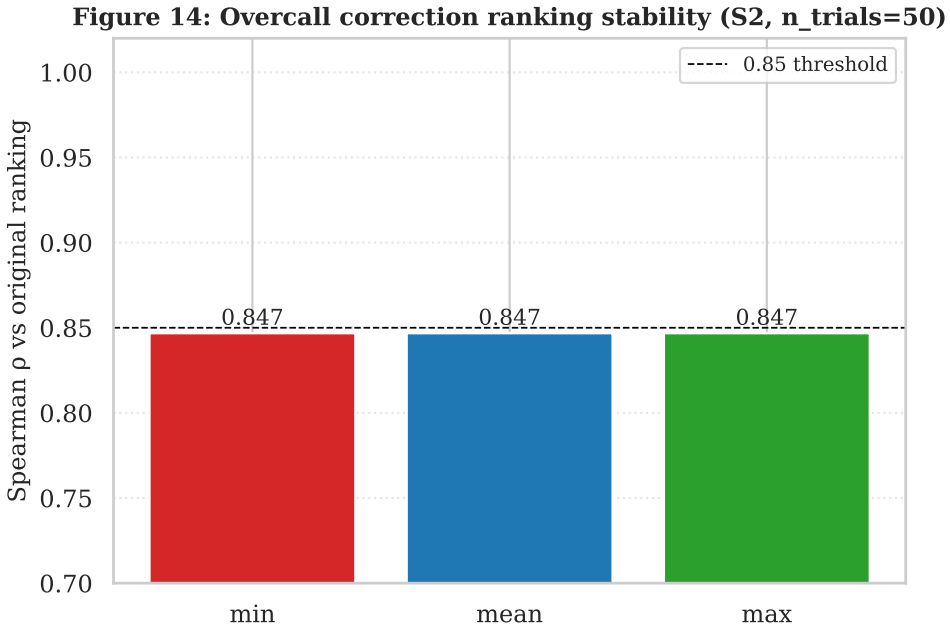

Figure 11: S2: ranking stability under overcall correction. Mean Spearman $\rho = 0.847$ across 50 bootstrap trials, narrowly below the pre-registered 0.85 threshold.

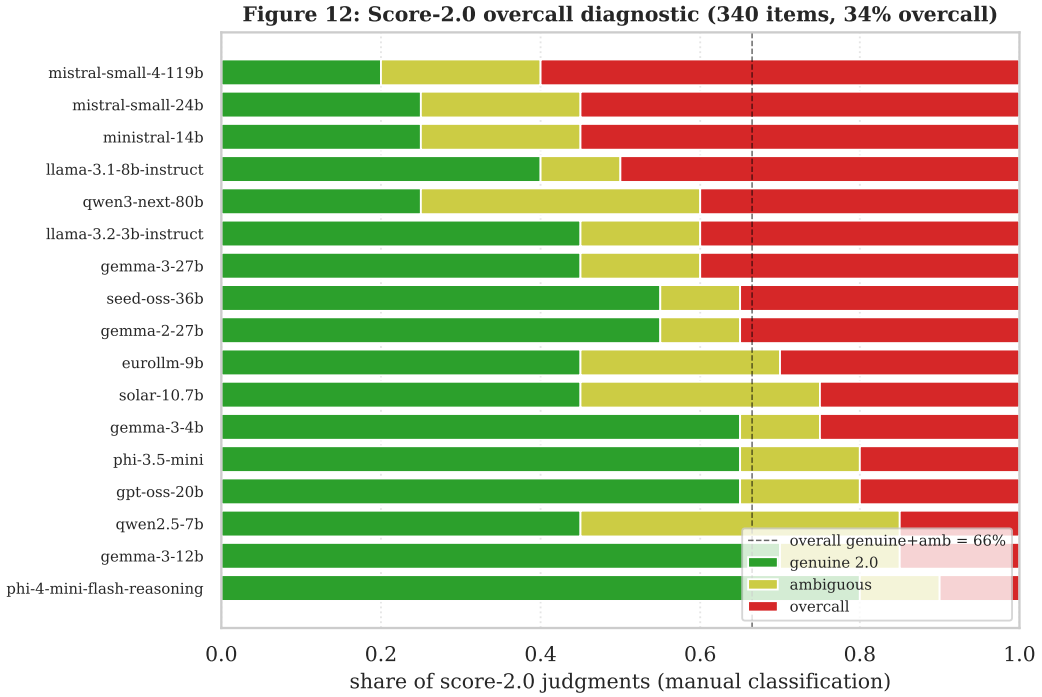

Figure 12: Per-model overcall breakdown on the 340-item human-rated subset (Appendix L).